

Centro de Ciencias Básicas

Ingeniería en Sistemas Computacionales

Programación de Sistemas Web

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ISC 5-A

Project proposal: MatchupIQ

Project Name MatchupIQ: Statistical Props Analysis & Odds Comparison Platform for NFL and Liga MX

Team Members

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GitHub Repository Link: https://github.com/leonardo-9206/MatchupIQ

Technology Stack:

Frontend:

HTML5: Semantic structuring for clean, accessible web pages.

CSS3 & Tailwind CSS: Responsive styling, modern dark mode, Flexbox and Grid layouts.

JavaScript: Dynamic DOM manipulation, event handling, and asynchronous data fetching using Fetch API.

Backend:

PHP 8: Server-side logic structured with the MVC (Model-View-Controller) architecture.

RESTful Endpoints: Controllers handling GET and POST requests returning structured JSON responses.

Security: Native session management, password hashing (password_hash), input sanitization, and parameterized SQL queries.

Database:

MySQL: Relational database normalized to 3NF to store users, sports, teams, players, prop metrics, bookmaker odds, and tracking logs.

Libraries and Dependencies:

Tailwind CSS: Utility-first framework for UI styling.

Chart.js / ApexCharts: Interactive data visualization for player statistical trends.

Lucide Icons / FontAwesome: Clean vector icons for interface components.

PHP PDO: Database abstraction layer for secure SQL operations with prepared statements.

cURL / Guzzle: Backend HTTP client to consume external sports APIs.

The Odds API: External service for live bookmaker odds covering NFL and Liga MX.

Project Description:

What problem the project aims to solve:

Sports enthusiasts and analysts face a lot of information fragmentation when researching player performance bets (props) for leagues like the NFL (passing yards, rushing attempts) and Liga MX (shots on target, goals), and currently, the necessary data is scattered all over the place:

Player stats and game logs are on sports media websites.

Defensive rankings and matchup indicators live on separate analytical portals.

Live odds and betting lines are locked inside individual sportsbooks like Playdoit, Draftea, and Caliente.

This forces people to manually open a dozen browser tabs, copy numbers, and cross-reference everything by hand, because of this, it is easy to miss obvious statistical discrepancies, like when a sportsbook sets a low line on a player facing a weak defense.

Why this problem is important?:

Evaluating sports props efficiently requires quick and reliable data aggregation. Doing this manually takes too much time, leads to human error, and makes it hard to place well-informed, data-driven picks before lines move. MatchupIQ solves this by providing a single, centralized tool that pulls this information together and calculates statistical value automatically, saving time and helping users make objective decisions.

How the project will solve it and Key Features:

MatchupIQ integrates data ingestion, analytical algorithms, and clear visualizations into one responsive web platform:

Centralized Dashboard: A unified interface showing weekly NFL and Liga MX matchups alongside opponent defensive rankings.

Value Edge Algorithm: Backend logic that compares player season averages against defensive weaknesses to highlight high-value lines automatically.

Multi-Bookmaker Odds Comparator: Side-by-side tables showing live betting lines across different bookmakers.

Dynamic AJAX Filtering: Instant searching and filtering by sport, team, position, or edge value without reloading the page.

User Bet Tracker (Full CRUD): User accounts where people can save specific props, add custom analysis notes, and track their results over time.

Admin Panel (Full CRUD): Restricted backend control panel to manage teams, player rosters, defensive metrics, and system data.